

Task Manager REST API — Setup Progress

Java + Spring Boot + MySQL + JWT + Git + GitHub + Codex

Project Goal

Build a resume-ready Task Manager REST API using Java and Spring Boot. The project will include user registration/login with JWT, task CRUD, validation, authorization, pagination/filtering, exception handling, unit tests, Git/GitHub, and Codex-assisted development.

Agreed Learning Approach

Go slowly and complete one step before moving to the next. For each stage: do the setup, understand what it does, test it, use Git where appropriate, and only then continue. Codex will be used as a development assistant, but not to blindly generate the entire project.

Step-by-Step Progress So Far

Step	What we did	Result / Understanding
1	Decided on the development environment	Used VS Code instead of STS/Eclipse. The project will be developed with Java 21, Maven, Git, GitHub
2	Checked Java	Verified Java is installed and working. Java version: 21.0.10 LTS.
3	Checked Java compiler	Verified javac is installed and working. javac version: 21.0.10.
4	Checked Git	Verified Git is installed and working. Git version: 2.53.0.windows.1.
5	Checked Maven	Initially, mvn was not recognized by PowerShell, so Maven was not yet available through PATH.
6	Downloaded Maven	Downloaded Apache Maven 3.9.16 binary ZIP for Windows.
7	Extracted Maven	Extracted Maven to C:\Program Files\Apache\apache-maven-3.9.16. The bin, boot, conf, and lib folders
8	Configured MAVEN_HOME	Created the system variable MAVEN_HOME pointing to C:\Program Files\Apache\apache-maven-3.9
9	Configured Maven PATH	Added %MAVEN_HOME%\bin to the System variables Path.
10	Verified Maven installation	Confirmed Maven can locate mvn.cmd and Maven now runs successfully with Java 21 on Windows 1
11	Git setup planned	Next Git configuration is to set the Git user name and GitHub email before making the first project con

Important Commands Learned / Used

Check Java

```
java -version
```

Check Java compiler

```
javac -version
```

Check Maven

```
mvn -version
```

Check Git

```
git --version
```

Check Maven installation directly

```
Test-Path "$env:MAVEN_HOME\bin\mvn.cmd"
```

Check Maven environment variable

```
$env:MAVEN_HOME
```

Check Maven-related PATH entries

```
$env:Path -split ';' | Where-Object { $_ -like '*Maven*' -or $_ -like '*maven*' }
```

Current Environment

Component	Status
Windows	Windows 11, 64-bit
Java	21.0.10 LTS
javac	21.0.10
Maven	Apache Maven 3.9.16 — working
Git	2.53.0.windows.1 — working
VS Code	Selected development environment
Spring Boot project	Not created yet
GitHub repository	Not created yet
Codex workflow	Not started yet

Next Step — Not Started Yet

The next step is to configure/check the Java and Spring Boot extensions in VS Code. After that, we will create the Spring Boot project directly in VS Code, initialize Git, make the first commit, and later connect the repository to GitHub.

Planned Project Build Order

- VS Code + Spring Boot extensions
- Create Spring Boot project
- Initialize Git and make the first commit
- Create/connect GitHub repository
- Configure MySQL and application.properties
- User entity and repository
- Task entity and repository
- DTOs and validation
- User registration and password encryption

- JWT authentication and Spring Security
- Task CRUD with user ownership
- Pagination and filtering
- Global exception handling
- JUnit + Mockito service tests
- Postman API testing
- Swagger/OpenAPI documentation
- README and GitHub cleanup
- Codex-assisted review, debugging, and refactoring

Learning rule: do not move to the next step until the current step is working and understood.